

MGTECON 603 - Questions(Wk 7)

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This is a document for the questions I had on the seventh week of the course.

1 Slides 11 More Regression Stuff

1.1 What exactly is the OVB logic?

OMITED VARIABLE BIAS

- Consider the regression

$$Y_i = \beta_0 + \beta_1 W_i X_i + \beta_2 W_i (1 - X_i) + \beta_3 X_i + \varepsilon_i$$

- Consider the regression:

$$Y_i = \theta_0 + \theta_1 (W_i - \delta_0 - \delta_1 X_i) + \varepsilon_i$$

- Then by omitted variable bias, in the population,

$$\theta_1 = \beta_1 \frac{\mathbb{E}[W_i X_i (W_i - \delta_0 - \delta_1 X_i)]}{\mathbb{E}[(W_i - \delta_0 - \delta_1 X_i)^2]} + \beta_2 \frac{\mathbb{E}[W_i (1 - X_i) (W_i - \delta_0 - \delta_1 X_i)]}{\mathbb{E}[(W_i - \delta_0 - \delta_1 X_i)^2]} + \beta_3 \frac{\mathbb{E}[X_i (W_i - \delta_0 - \delta_1 X_i)]}{\mathbb{E}[(W_i - \delta_0 - \delta_1 X_i)^2]}$$

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Figure 1.1: p.12 of the slides

Figure 1.1 shows the slides 11.12. In this figure we derived the relationship between the parameters of two regression models, using the so-called OVB logic. The way I understood the logic is that, we start with the first regression model and consider that the true DGP. In the second regression, we have the variable $W_i - \delta_0 - \delta_1 X_i$ as the univariate regressor. Denote this regressor by Z_i . We can decompose the correlation between this variable and the outcome Y_i into three parts, or channels:

- the channel of $W_i X_i$: computed by the product of the direct effect of $W_i X_i$ on Y_i , which is β_1 , and the correlation between $W_i X_i$ and Z_i , which is the cross-moment in the slide.
- the channel of $W_i(1 - X_i)$: computed by the product of the direct effect of $W_i(1 - X_i)$ on Y_i , which is β_2 , and the correlation between $W_i(1 - X_i)$ and Z_i , which is the cross-moment in the slide.
- the channel of X_i : computed by the product of the direct effect of X_i on Y_i , which is β_3 , and the correlation between X_i and Z_i , which is the cross-moment in the slide.

This kind of logic helped me “memorize” the OVB logic and the formula; however, I still don’t understand how equation in the slide makes sense. This doesn’t look like the FWL theorem to me, because in that case, we were comparing two regression models where one has the entire set of covariates, and the other has some omitted variables. That doesn’t directly translate to the equations I see in the slide.

2 Slides 13 Lasso Regression

This question relates to the figure comparing the coefficients of the lasso and the ridge regression:

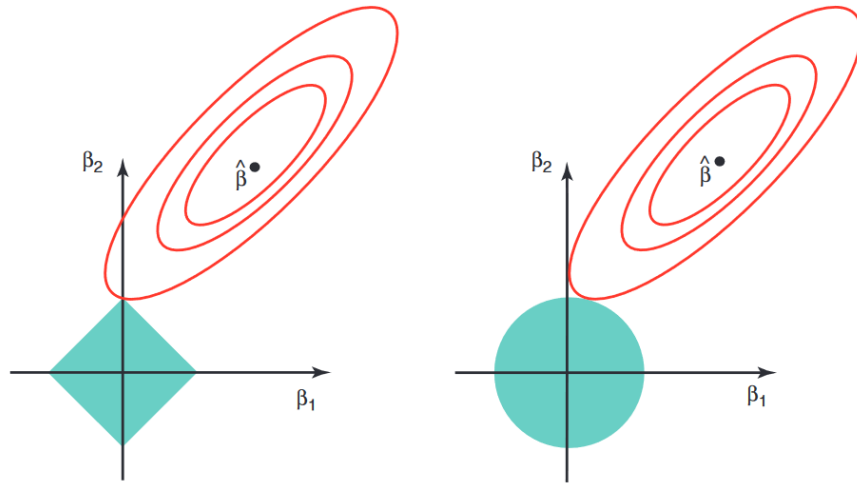


Figure 2.1: LASSO vs. Ridge Regression

I’m more interested in the shape of the ovals. The oval here looks like it’s stretched along the $y = x + k$ line, but Guido drew an oval stretched along the $y = -x + k$ line (sort of stretched from NW to SE) when we have multicollinear variables. I wonder how to understand the shape of the ovals. As an update from Nov 12, I asked this question to Guido, but I didn’t fully understand his answer yet. He said that the oval becomes a circle if the covariates are fully independent, but becomes more oval if we have collinearity. But my question is more related to ‘which’ direction the oval would be stretched to.

One intuition I had about these figures before was comparing them to the utility curve and the budget constraint in the consumer theory – that way, I could kind of think of the ovals as the indifference curves. I wonder if we can apply a similar intuition and compare the collinear case to the perfect substitute case in consumer theory – we have a flatter tangent of the oval(indifference curve) when we have substitutable variables(goods).

3 Slides 14 Clustering

RECAP

- Best linear predictor set up
- Even weaker assumptions: moments of X_i and ε_i exist,

$$(\varepsilon_i, X_i) \text{ iid}, \quad \mathbb{E}[\varepsilon_i X_i] = 0$$

Then:

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} \mathcal{N}\left(0, \left(\mathbb{E}[X_i X_i^\top]\right)^{-1} \left(\mathbb{E}[\varepsilon_i^2 X_i X_i^\top]\right) \left(\mathbb{E}[X_i X_i^\top]\right)^{-1}\right)$$

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Figure 3.1: Clustering

In the slide above(fig. 3.1), it assumes that the error term and the covariates are iid distributed. Is this a typo? To write the true asymptotic variance in the way it is written down on the slide, isn't 'independence' sufficient for this case, and the error terms need not have the exact same distribution?